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Comparing electro- and mechano-myographic muscle activation patterns in self-paced pediatric gait

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ABSTRACT

Electromyography (EMG) is the standard modality for measuring muscle activity. However, the convenience and availability of low-cost accelerometer-based wearables makes mechanomyography (MMG) an increasingly attractive alternative modality for clinical applications. Literature to date has demonstrated a strong association between EMG and MMG temporal alignment in isometric and isokinetic contractions. However, the EMG-MMG relationship has not been studied in gait. In this study, the concurrence of EMG- and MMG-detected contractions in the tibialis anterior, lateral gastrocnemius, vastus lateralis, and biceps femoris muscles were investigated in children during self-paced gait. Furthermore, the distribution of signal power over the gait cycle was statistically compared between EMG-MMG modalities. With EMG as the reference, muscular contractions were detected based on MMG with balanced accuracies between 88 and 94% for all muscles except the gastrocnemius. MMG signal power differed from that of EMG during certain phases of the gait cycle in all muscles except the biceps femoris. These timing and power distribution differences between the two modalities may in part be related to muscle fascicle length changes that are unique to muscle motion during gait. Our findings suggest that the relationship between EMG and MMG appears to be more complex during gait than in isometric and isokinetic contractions.

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1. Introduction

Knowledge of muscle activity during dynamic activities such as gait is useful for the elucidation of motor control and learning strategies (Krueger et al., 2014), assessment of motor disabilities, monitoring the progress of rehabilitation, development of neuroprosthesis (Cè et al., 2015b; McLaren et al., 2016), and informing clinical decision-making (Hossain et al., 2014). In particular, a comprehensive picture of muscle function can be derived from continuous *in situ* recordings of muscle activity, as enabled through wearable technologies, or "wearables." Wearables consist of small, unobtrusive sensors attached to the body or to clothing that monitor physiological and behavioral signals over extended periods of time. For example, wearables utilizing movement sensors have been helpful in measuring gait dynamics, joint kinematics, and the effectiveness of at-home rehabilitation efforts (McLaren et al., 2016; Steins et al., 2014). Furthermore, these smaller-sized wearables would be most useful in studying the smaller muscles in a pediatric population.

Surface electromyography (EMG) records electrophysiological impulses over the muscle belly during muscle contraction, and is the gold standard for measuring muscle activity (Cè et al., 2015b). EMG applications historically include clinical diagnosis and rehabilitation monitoring, and more recently, postural biofeedback, and control of human machine interfaces, such as communication devices and video games (Mazumder and Kundu, 2012; McLaren et al., 2016; Merletti and Farina, 2016). An alternative to measuring the electrical impulses at the onset of muscle activity is to measure the mechanical force transmission of muscle fibres via mechanomyography (MMG). MMG measures the lateral oscillations of active and passive parts of the series elastic component of the musculotendinous unit using various transducers, such as microphones or accelerometers (Cè et al., 2015b; Hossain et al., 2014; Krueger et al., 2014). MMG has been used to describe motor control strategies, in terms of motor unit summation (Orizio et al., 1996), firing pattern during fatigue (Orizio et al., 2003), and force during contractions (Madeleine et al., 2001).

Some studies have explored the temporal relationship between EMG and MMG in voluntary isometric and isokinetic dynamic

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contractions, reporting electromechanical delays between 20 and 125 ms during voluntary contractions (Blackburn et al., 2009; Yavuz et al., 2010). Additionally, lower limb studies in cycle ergometry have reported overlapping bursts of activity for simultaneously recorded EMG and MMG (Begovic et al., 2014). Consequently, a number of kinesiological and clinical studies have deployed MMG as a complementary signal to EMG in assessing postural balance and age-related muscle changes (Cè et al., 2015b; Hossain et al., 2014). While EMG signals are not known to exhibit differences between concentric and eccentric contractions, Jaskólska et al. (2004) reported contraction-specific MMG frequency and amplitude responses in the upper extremity agonist and antagonist muscles. Perry-Rana et al. (2003) discovered differences in the EMG and MMG responses among the vastus lateralis, rectus femoris and vastus medialis during maximal eccentric contractions while in an earlier study, Perry et al. (2001) hinted at an association between the metabolic and MMG characteristics of muscular contraction. Additionally, the relationship between EMG and MMG frequency spectral patterns is more complex in voluntary than in electrically induced contractions (Yoshitake and Moritani, 1999; Yoshitake et al., 2002). Collectively, these findings suggest that the relationship between MMG and EMG signals during gait may not be straightforward and that MMG may bear uniquely complementary information about the underlying muscle activity. To our knowledge, the temporal patterns of lower limb MMG and their correspondence to simultaneously recorded EMG activity have not been explored during gait.

The availability of accelerometer-based wearables makes MMG a feasible alternative to printed electrodes for EMG wearables (Matsuhisa et al., 2015). Furthermore, a systematic review examining the use of wearable accelerometer-based technology for neurological populations in the community highlighted the effectiveness of these technologies in distinguishing between typical and atypical mobility patterns (Steins et al., 2014). Additionally, the smaller size of the accelerometer-based MMG sensor is particularly conducive to application in pediatric rehabilitation settings. Although some studies have deployed MMG alongside EMG or functional electrical stimulation in neuromuscular populations (Krueger et al., 2011; Marusiak et al., 2009a,b), there have been no studies specifically focusing on MMG patterns in pediatric populations. In particular, the level of temporal concordance between EMG and MMG signals of the lower limbs during self-paced gait in children remains unknown.

In this study, we measured MMG of the lower limb muscles simultaneously with EMG during self-paced gait in a typically developing pediatric population to evaluate the temporal patterns of MMG activity during the gait cycle relative to EMG, and gain insight into the corresponding muscle-specific, accelerometer-based mechanical activity.

2. Methods

2.1. Participants

We recruited 20 typically developing pediatric participants (5 males and 15 females) between the ages of 8 and 18 (mean 14.3 ± 3.2, range 8–18) years with an average height of 158.8 ± 11.6 cm and an average weight of 55.5 ± 19.2 kilograms. This age group was chosen to ensure that participants' gait patterns had developed into mature, adult-like patterns (Sutherland, 1997). Participants reported no known neurological disorders, or musculoskeletal injuries (current or previous) limiting their walking. Participants wore athletic shoes and shorts during the study. The Holland Bloorview hospital and University of Toronto Research Ethics Boards approved the study protocol, and all participants provided informed written consent.

2.2. Instrumentation

Wireless surface electromyography (EMG) sensors (Trigno by Delsys Inc, Boston, MA, USA) were fastened with double-sided tape over the muscle belly of each of the following lower limb muscles: tibialis anterior (TA), lateral gastrocnemius (LG), vastus lateralis (VL), and biceps femoris (BF). MMG data were collected using triaxial accelerometers (ADXL337, Analog Devices Inc, Norwood, MA). The accelerometers were positioned on the muscle bellies of the same muscles, usually 3 cm proximal to the EMG sensor (Fig. 1). For participants who were less than 145 cm in height and under 33 kg in weight, we situated the accelerometer 2 cm from the EMG sensor, to ensure proximity to the muscle belly and to avoid the corresponding tendon. For each muscle, the accelerometer was oriented such that the z-axis was perpendicular to the longitudinal axis of the muscle. An ultra-thin, force-sensitive resistor (FSR) (FSR 406, Interlink Electronics, Shenzhen, China) was inserted inside the participant's shoe to record heel strike for each step (Sejdić et al., 2012). All data were collected bilaterally.

Each sensor was connected to a data logging system housed inside a backpack (total mass: 1.83 kg) worn by the participant. All data were acquired via a custom-made LabVIEW program at a sampling rate of 1 kHz for MMG and 2 kHz for EMG.

2.3. Data collection

Prior to the walking trial, we collected 20 s of baseline MMG and EMG during quiet standing. Participants were then instructed to walk continuously at a self-selected pace on an obstacle-free, rectangular-shaped, well-lit indoor gym track (5 m × 8 m with linoleum flooring). All participants walked counter-clockwise around the track continuously for 15-min. Participants did not complain of fatigue or discomfort from wearing the sensors and backpack, and all participants were able to complete the full 15-min of walking.

2.4. Signal processing

To extract MMG content from the accelerometry signals, the z-component of the signal was bandpass filtered (5th order Butterworth filter) between 5 and 50 Hz and full-wave rectified. EMG data were bandpass filtered (4th order Butterworth filter) between 30 and 500 Hz, full-wave rectified, and down-sampled to 1 kHz (Reaz et al., 2006). Both MMG and EMG signals were smoothed using a moving window average of 101 samples and separately normalized to the interval [0,1]. All data analysis was carried out via a custom-designed MATLAB program. We extracted 2 min of data starting at the 2-min mark of each participant's 15 min recording for subsequent EMG-MMG temporal validation analyses.

2.5. Co-incident EMG-MMG activity

The percent of overlap between EMG and MMG within a given window of observation was defined as the fraction of time in which normalized versions of both signals exceeded an amplitude threshold (described below), i.e., were "active". Windows within which the overlap between MMG and EMG active regions was at least 80% were identified as segments of co-incident activation.

To identify muscle-specific co-incident MMG and EMG activity, we found the window size and the signal amplitude threshold that together maximized the agreement between the active regions of the two signals. In other words, using EMG activity as the reference, we determined a window length-amplitude threshold

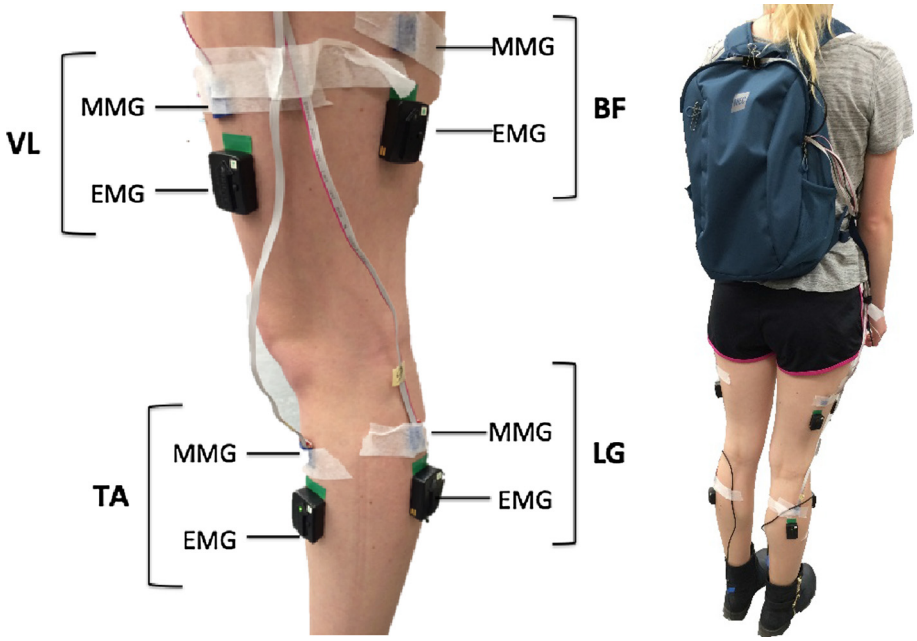

Fig. 1. EMG and MMG sensors attached to the participants' muscles (TA = tibialis anterior, LG = lateral gastrocnemius, VL = Vastus lateralis, BF = biceps Femoris) on one leg (left) and the backpack worn by the participant containing the MMG data board and tablet (right).

pairing that maximized the balanced accuracy of detecting MMG activity. The exhaustive search for this optimal pairing was performed over a discrete grid of amplitude thresholds (0.05–0.5 in increments of 0.05) and moving window sizes (100–1500 ms in increments of 50). Balanced accuracy was defined as the arithmetic mean of sensitivity (true positive rate) and specificity (true negative rate). A true positive occurred when MMG activity was present given corresponding EMG activity whereas a false negative indicated an active EMG segment with no corresponding MMG activity. A true negative was tallied when both EMG and MMG activities were absent within a window of time, while a false positive was an instance of active MMG with no concurrent EMG activity (Fig. 2). Balanced accuracies were compared across muscles with a one-way analysis-of-variance using RStudio for R (Boston, USA). One-way analyses-of-variance were also invoked to evaluate across-muscle differences in window length and amplitude thresholds.

2.6. MMG stride characterization

For the 15-min of self-paced gait, EMG and MMG signals were segmented based on the heel strikes as identified in the FSR signals. Fifty steps from the beginning and end of the session were discarded to account for familiarization and termination effects, respectively, and steps <400 ms and >2000 ms were also discarded to account for missteps or stops during the trial. With this procedure, no more than 15 steps were eliminated for a given participant. The length of each stride was then normalized to 2 s, and

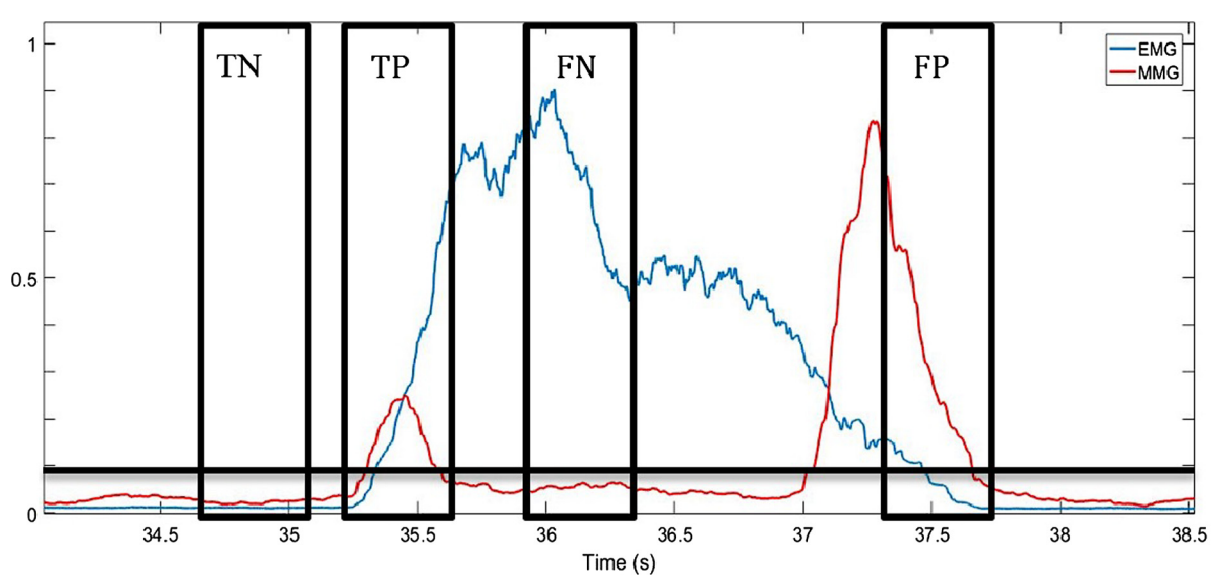

Fig. 2. Determining coincident activity of EMG and MMG signals using an amplitude threshold (black line) and moving window (box). Examples of true negative (TN), true positive (TP), false negative (FN) and false positive (FP) cases are illustrated.

the power of the EMG and MMG signals was separately calculated within each 20% division of the gait cycle. Muscle-specific differences in signal power between EMG and MMG signals over the gait cycle were assessed using a repeated measures two-way analysis of variance with measurement modality (EMG; MMG) and gait phase (0–20; 20–40; 40–60; 60–80; 80–100%) as independent factors. Significant differences in power were further investigated via post hoc pairwise *t*-tests with a Bonferroni adjustment for multiple comparisons. The statistical analysis was performed using RStudio. The above analyses (comparing balanced accuracies across muscles and the effects of measurement modality and gait phase on signal power) were replicated in subgroups of gender, age (8–12 years and 13–18 years) and body size (<60 kg and >60 kg).

3. Results

Balanced accuracies averaged across participants for different window lengths and amplitude thresholds are shown in Fig. 3. High accuracy suggesting strong coincidence between EMG and MMG activities was seen for different amplitude threshold-window size pairings for the TA, VL, and BF muscle sites. However, the maximum balanced accuracy for the LG muscle sites of both right and left legs were significantly lower than that of all other muscle sites ($p < 0.001$), indicating low coincidence between MMG and EMG-demarcated activity. This finding is corroborated when examining the maximum balanced accuracy achievable for each muscle on each side, as shown in Table 1. No significant differences were seen among muscle sites for window sizes ($p = 0.74$) or threshold amplitudes ($p = 0.48$) for maximum balanced accuracies across participants.

During the gait cycle (Fig. 4), coincident activity between EMG and MMG for the TA, VL, and BF was observed primarily after heel strike (0–20%) and through swing phase (60–100%). Despite these similar patterns, there was a significant interaction between modality (EMG and MMG) and gait cycle division for the TA, LG, and VL muscles bilaterally, and the BF for the left side only ($p < 0.001$), which indicates differences between EMG and MMG at different subintervals of the gait cycle. For the TA muscle, subsequent paired *t*-tests for each division of the gait cycle revealed significantly higher MMG than EMG power mid-cycle ($p < 0.001$) and significantly lower MMG at the end of gait cycle ($p < 0.05$). This trend was more prominent on the right versus left side (Fig. 5). For the LG muscles bilaterally, we observed significantly higher MMG than EMG power following heel strike and during swing phase (60–100%) ($p < 0.01$), and significantly lower MMG power mid-cycle where we observed the majority of the EMG activity ($p < 0.001$). Although the VL EMG and MMG patterns appeared similar for the given participant in Fig. 4, the mean EMG and MMG signal powers across participants differed mid-cycle for the right side. However, for the left-sided VL, we observed significantly higher EMG activity at the beginning of stance, and significantly higher MMG activity at the beginning of swing phase ($p < 0.05$) (Fig. 5). A similar trend was seen between left-side VL and BF muscle; however, the BF shows significantly higher MMG activity following heel strike ($p < 0.01$). There was no significant interaction between modality and gait cycle division, and there was no significant effect of modality on power of the right-sided BF muscle only. Findings from the subgroup analyses by gender, age and body size were consistent with those derived from the entire sample.

4. Discussion

4.1. Coincident MMG and EMG activity

When comparing the processed MMG and EMG for the shank and thigh agonist-antagonist muscles, we observed bursts of MMG activity that corresponded to EMG activity. This MMG-EMG correspondence was quantitatively verified by the high balanced accuracies for the TA, VL, and BF muscles, and corroborates the qualitative reports of previous dynamic studies (Perry et al.,

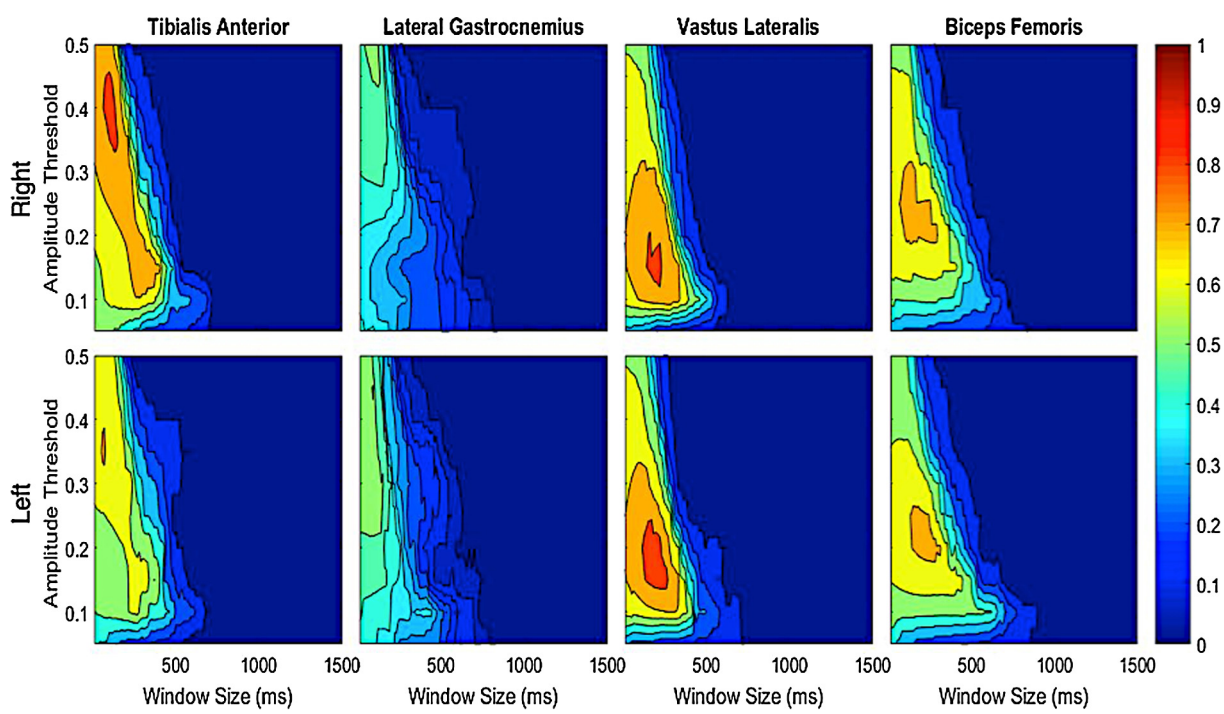

Fig. 3. Balanced accuracies (BACC) averaged across participants, showing degree of EMG and MMG signal alignment as a function of amplitude threshold (vertical axis) and window size (horizontal axis) for each muscle site on the right (top row) and left (bottom row) legs. BACC values closer to 1 (red) indicate greater coincident activity between EMG and MMG. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Table 1
Maximum co-incident activity between EMG-MMG signals by muscle and side (right vs. left leg) as measured by maximum balanced accuracies (first row). The subsequent rows report the corresponding optimal values of window sizes and amplitude thresholds. Values shown are mean and standard deviation across all participants.

	Tibialis anterior		Lateral gastrocnemius		Vastus lateralis		Biceps femoris	
	Right	Left	Right	Left	Right	Left	Right	Left
Maximum balanced accuracy	0.94 ± 0.15	0.93 ± 0.14	0.61 ± 0.19	0.71 ± 0.21	0.93 ± 0.15	0.92 ± 0.14	0.89 ± 0.18	0.89 ± 0.18
Window size (ms)	414 ± 147	435 ± 138	408 ± 169	398 ± 141	394 ± 108	420 ± 110	405 ± 152	471 ± 167
Threshold (mV)	0.21 ± 0.11	0.23 ± 0.16	0.21 ± 0.13	0.16 ± 0.08	0.18 ± 0.11	0.17 ± 0.09	0.17 ± 0.11	0.20 ± 0.13

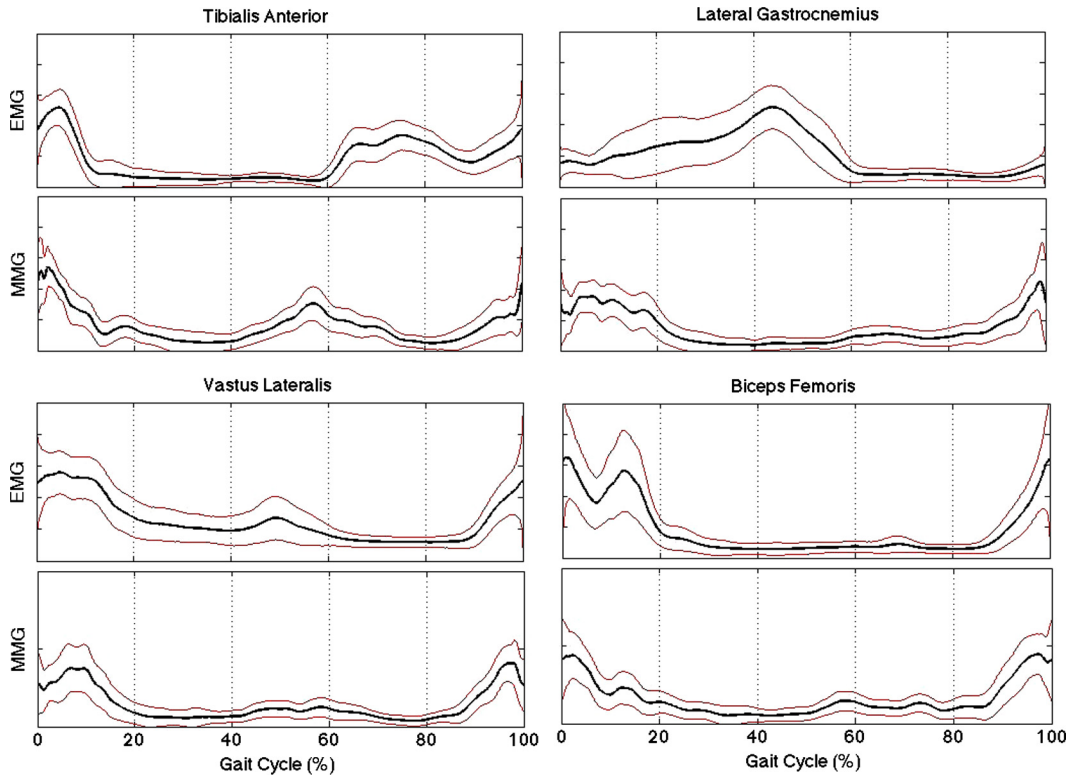

Fig. 4. For visualization purposes, the average of the normalized EMG and MMG signals were calculated across all steps for each participant. The typical activity patterns are shown for one participant (p28), showing the mean (black) with standard deviation (red) for all muscles of the right leg. The gait cycle starts at heel strike, and swing phase typically begins at 60% of the gait cycle. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

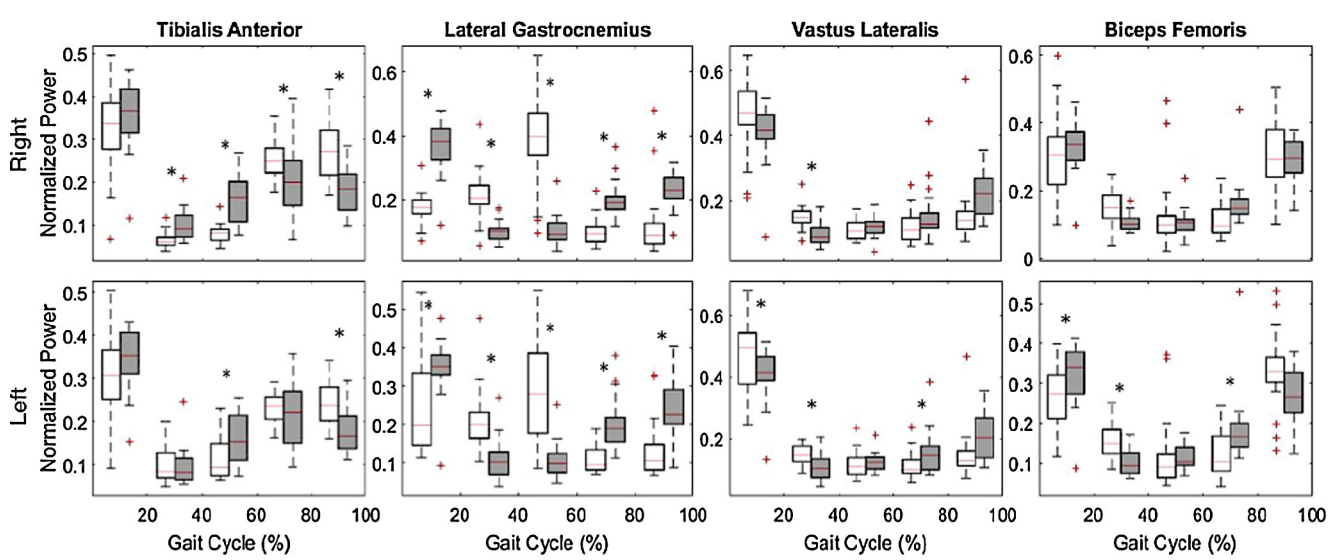

Fig. 5. Mean EMG (clear box) and MMG (shaded box) signal power across all participants for the right (top row) and left (bottom row) legs. The asterisks (*) specify significant differences between EMG and MMG power within an interval of the gait cycle divisions.

2001; Shinohara et al., 1997). For example, in studying the relationship between leg extensor activity and work load during cycle ergometry, Shinohara et al. remarked that the shapes of the rectified MMG "roughly approximated" those of the EMG signals but with "some" time delay (Shinohara et al., 1997). Additionally, Shinohara et al. noted the presence of MMG signal activity during the "non-contraction phase" of the muscle where no corresponding EMG was visible, attributing this discrepancy to vibrations propagating from the antagonist muscles (Shinohara et al., 1997). This antagonist muscle "noise" may in part account for the lower balanced accuracies in our study. In studying the relationship between EMG, MMG, and exerted power in incremental cycle ergometry, Perry et al. (2001) reported 10-s of raw EMG and MMG activity, where it appears that MMG activity generally aligns with EMG activity, with limited MMG activity between contractions. Although they did not discuss this relationship, they did conclude that patterns of MMG amplitude may be more useful than EMG for examining power output during continuous, dynamic tasks (Perry et al., 2001).

4.2. Discrepant MMG and EMG activity

Interestingly, for the LG muscle, we found a large discrepancy between the EMG and MMG signals. The accuracy of detecting active regions was only about 65% and the distribution of power over the gait cycle was visibly different between modalities (Fig. 4). The EMG power was concentrated around the 50% mark of the gait cycle, i.e., when the shank is rolling over the foot and the ankle is plantar flexing to maintain forward motion, which is the typical profile of EMG activity during self-paced gait (Perry and Davids, 1992). In our study, we observed a large peak in MMG power following heel strike, then again around 60% of the gait cycle, and a third peak following toe off and before heel strike (Fig. 4). The observed MMG activity before and after heel strike has not been noted in previous EMG/MMG studies (Perry et al., 2001; Shinohara et al., 1997). Further, the observed MMG-EMG temporal misalignment is beyond the known electromechanical delay in voluntary, dynamic studies (Begovic et al., 2014; Blackburn et al., 2009). The MMG signal for the LG muscle may in fact comprise vibrations from the passive movement of surrounding non-muscular, soft tissues, and contraction of the antagonist muscles. The latter is suggested by the peaks we observed in the MMG power of the LG at the end of swing phase and after heel strike, corresponding to periods of electrical activity of the TA (Fig. 5). Previous research examining EMG and MMG during cycle ergometry concluded that some of the MMG amplitude may be occurring from incomplete muscle relaxation and passive muscle fibre movement (Shinohara et al., 1997), which may explain some of the discrepancies in our study between EMG and MMG signal power during varying phases of the gait cycle. That agonist and antagonist co-activation manifested in MMG was also previously noted, albeit in an upper extremity study following eccentric exercise, where the authors argued that such co-contraction provides joint stability (Jaskólski et al., 2007).

4.3. Distribution of MMG signal power over gait cycle

MMG reflects the lateral oscillations of active and passive parts of the series elastic component of the musculotendinous complex (MTC) and in particular, the muscle fibres that create mechanical force (Cè et al., 2015b; Hossain et al., 2014; Krueger et al., 2014). Based on the force-velocity relationship, force is optimal at a certain muscle length during isometric and slow concentric actions; however, that relationship is more complex in dynamic actions (Chleboun et al., 2007). The muscle-tendon complex (MTC) is made up of muscle fibres and the tendons attaching them to bones. Muscle fibres transmit force to and interact with the tendons given tendon compliance. Therefore, it is important to understand the relationship between the lengths of both muscle fibre and tendon during dynamic activities. In their measurements of fascicle length change and EMG activity of the TA and VL during walking, Chleboun et al. (2007) found significant fascicle length change and bursts of EMG activity in the TA between 60 and 80% of the gait cycle. Similarly, for the TA muscle, we found an aggregation of MMG signal power during the first 20% of the gait cycle and again around 60%. Furthermore, in the VL, Chleboun et al. (2007) observed significant length change and EMG activity at the end of the gait cycle (75–100%); however, fascicle length did not exhibit significant length change in the first or second portions of the gait cycle. Similarly, in our study, we observed an accumulation of MMG signal power for the VL muscle in the initial and final 20% of the gait cycle (Figs. 4 and 5). Thus, our observations of MMG signal power distribution over the gait cycle, can in part be understood in terms of underlying fascicle length changes.

4.4. Differences between MMG and EMG signal power distribution over the gait cycle

We found lower coincident EMG and MMG activation for the LG, and significant differences between the modalities in the various stages of the gait cycle (Fig. 5). Prior to the swing phase, we observed substantial electrical activity as the gastrocnemius activated to assist with push-off. In their study using ultrasonography to examine MTC during various dynamic movements, Fukunaga et al. (2002) showed that during the stance phase of walking, the medial gastrocnemius (MG) muscle fibres maintained constant length while the tendon stretched, whereas during push-off, both MTC and tendon shortened rapidly. In our study, we saw high EMG power at the end of stance phase (40–60%) with low corresponding MMG power (Fig. 5). At this point of the gait cycle, the gastrocnemius muscle is contracting nearly isometrically (Fukunaga et al., 2002), where we typically see lower peak-to-peak MMG amplitudes in comparison to concentric and eccentric components of contraction (Cè et al., 2015b; Madeleine et al., 2001). Thus, we should expect lower MMG activity during this period of electrical activity. Furthermore, the Achilles tendon lengthens during stance and recoils in push-off (Fukunaga et al., 2001), potentially contributing to the MMG signal at these times. Additionally, we saw differences in modality-specific power between left and right legs for the TA, VL and BF muscles, which may be related to participants walking with the left leg always on the inside of the oval track. Although we know that EMG muscle activity must be adjusted in order to make a turn during gait, overall gait patterns remain unchanged (Hase and Stein, 1999). Nevertheless, these findings suggest that the relationship between EMG, MMG and the associated temporal delay may be more complex during dynamic movements and future studies ought to consider muscle-specific MTC length changes in interpreting lower limb MMG during gait.

4.5. Limitations and future work

MMG signal characteristics may vary between adjacent muscles, specifically medial versus lateral gastrocnemius. Although some studies show comparable EMG activity between the LG and MG heads when recording different speeds of walking and running (Cappellini et al., 2006; Ivanenko et al., 2004; Perry and Davids, 1992), other studies focus only on MG given its larger muscle belly and greater activation during self-paced walking (Granata et al., 2005). Furthermore, differences in LG and MG muscle fibre architecture, such as muscle fascicle length, their insertion angle, and muscle thickness, suggests that there may be differences in the

MMG signals of these two muscles (Cè et al., 2015a; Koryak, 2008). We did not see any effects of gender, age, or body size on MMG power distribution and timing. However, future research should investigate potential effects of age and body size on MMG frequency responses in dynamic contractions (Jaskólska et al., 2004) in developing and advanced aging populations. Future studies should also investigate MMG differences between the MG and LG muscles as manifested during gait.

Despite the aforementioned inter-contraction noise, the average MMG signal for the TA muscle revealed two distinct peaks corresponding to one EMG burst, one at the start and the other at the end of the contraction (Fig. 4). Beck et al. (2008) observed, via wavelet transform, MMG spectral differences between the concentric and eccentric components of isokinetic contractions. Thus, our finding of dual MMG peaks, may suggest that spectral filtering of MMG has potential to differentiate between concentric and eccentric components of the contraction in gait. This would be particularly important given that EMG spectral analysis does not distinguish between concentric and eccentric contractions (Jaskólska et al., 2007; Linnamo et al., 2002; Madeleine et al., 2001). Wavelet-based MMG analysis for automatic detection of concentric and eccentric contractions during gait thus merits further exploration.

5. Conclusions

During pediatric gait, electro- and mechano-myographic activation patterns appear to be temporally aligned for the tibialis anterior, vastus lateralis and biceps femoris muscles of both legs. However, for the lateral gastrocnemius muscle, the EMG and MMG activations exhibit large temporal discrepancies, well beyond that attributable to electromechanical delay. Passive vibrations of nearby tissues and vibrations of the antagonist muscles may contribute to the observed EMG and MMG activation offsets at the LG muscle. Differences between EMG and MMG signal power distributions over the gait cycle may be related to fascicle elongation/shortening and muscle-specific musculotendinous complex length changes during gait. The reported discrepancies between EMG and MMG temporal distributions of signal power suggest a complementary role for MMG in identifying and tracking electromechanical changes in musculotendinous behaviour due, for example, to injury, disease or training.

Conflict of interest

The authors declared that there is no conflict of interest.

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